Crime Hotspot Prediction Using Spatio-Temporal Machine Learning Models

Machine Learning Project Report

Group 7

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1 Introduction

Crime analysis and prediction have become important applications of data science and machine learning. With the rapid growth of urban populations and the increasing complexity of criminal activities, law enforcement agencies require advanced analytical tools to monitor crime patterns and respond effectively. Traditional crime analysis methods mainly rely on manual investigation and statistical summaries, which often fail to capture complex relationships present in large datasets.

Machine learning techniques provide a powerful alternative for analyzing crime data. These techniques can process large volumes of historical crime records and identify hidden patterns based on spatial and temporal characteristics. By analyzing factors such as crime location, time of occurrence, type of crime, and other contextual attributes, machine learning models can predict areas where crimes are more likely to occur.

Crime hotspot prediction focuses on identifying geographical regions that experience a high frequency of criminal activities. Detecting such areas in advance can assist police departments in deploying resources more efficiently and implementing preventive strategies. Instead of reacting after crimes occur, predictive systems allow authorities to take proactive measures to reduce crime rates.

In this project, a machine learning pipeline is developed to analyze historical crime datasets and predict potential crime hotspots. Several machine learning models are implemented and compared, including Random Forest, XGBoost, LightGBM, and CatBoost. These algorithms are widely used in predictive analytics due to their high accuracy and ability to handle complex datasets.

The main goal of this study is to analyze crime data, extract meaningful spatio-temporal features, and evaluate the effectiveness of different machine learning models in predicting crime hotspot regions.

1.1 Problem Statement

Identifying and predicting high-risk crime areas in advance is difficult using traditional analysis techniques. Manual crime analysis often leads to delayed responses and inefficient allocation of police resources. There is a need for intelligent predictive systems that can analyze historical crime data and accurately identify potential crime hotspots using machine learning models.

1.2 Objectives

The objectives of this project include:

- To analyze historical crime datasets and understand the patterns of criminal activities.
- To perform exploratory data analysis in order to visualize and interpret crime distributions.
- To extract relevant spatial and temporal features from the dataset.
- To implement multiple machine learning models for crime hotspot prediction.
- To evaluate and compare the performance of the implemented models using appropriate metrics.

2 Dataset Description

The dataset used in this project consists of crime records collected between the years 2020 and 2022. Each record represents a crime incident along with detailed information about the time, location, and type of crime.

The dataset contains several important attributes that help in analyzing crime patterns. These attributes include the date and time of the incident, the type of crime committed, geographical coordinates such as latitude and longitude, the community area in which the crime occurred, and whether an arrest was made.

Spatial features such as latitude and longitude provide information about the geographical location of the crime. These features are essential for identifying crime clusters and hotspot regions. Temporal features such as hour, day, and month allow us to analyze patterns related to

time, such as whether crimes occur more frequently during certain hours of the day or specific days of the week.

Before applying machine learning models, the dataset undergoes preprocessing to remove missing values and convert the date column into a suitable datetime format. Additional features such as hour, month, and day of the week are extracted from the date column to improve the predictive capability of the models.

3 Exploratory Data Analysis

Exploratory Data Analysis (EDA) is performed to understand the characteristics and distribution of the crime dataset. This step helps identify important patterns, trends, and relationships within the data.

3.1 Crime Distribution by Type

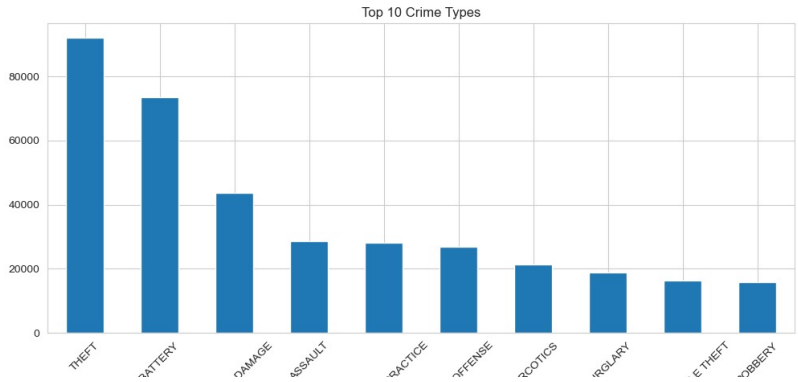

Figure 1: Top Crime Types

This figure shows the most frequently occurring crime types in the dataset. Understanding the distribution of crime categories helps identify which types of crimes are most common in the studied region.

3.2 Crime Count Distribution

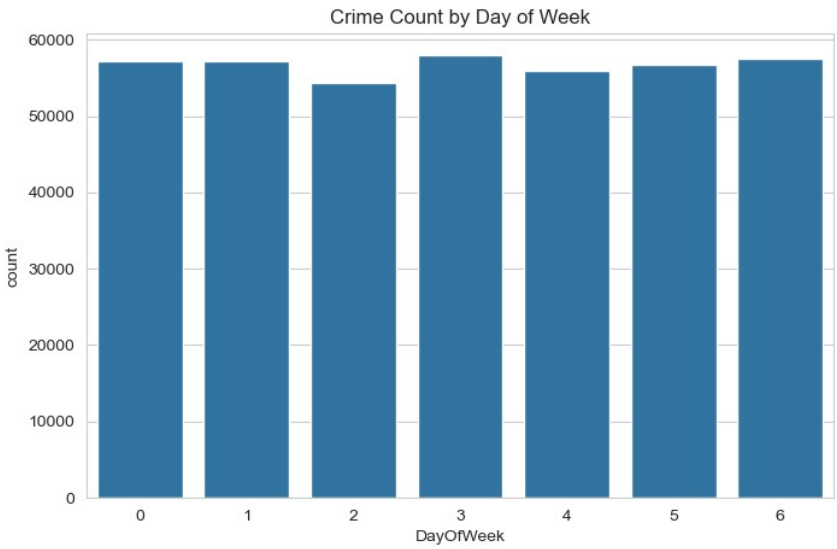

Figure 2: Crime Count Distribution

The crime count distribution provides an overview of how crimes are spread across the dataset. This visualization helps understand the relative frequency of crime incidents.

3.3 Crime Occurrence by Hour

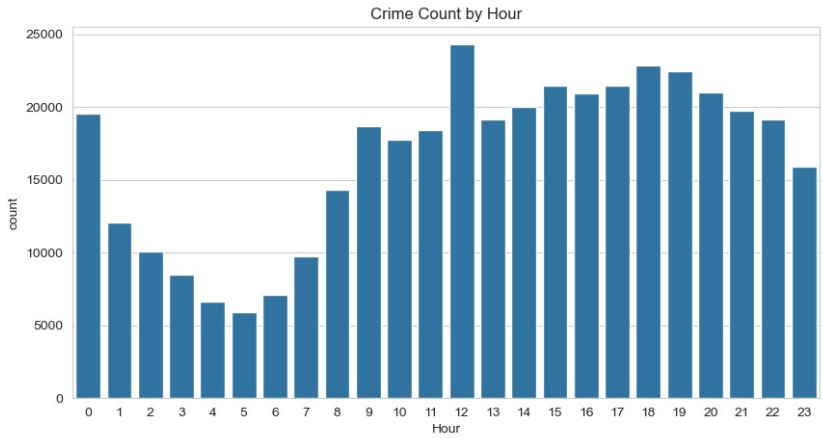

Figure 3: Crime Distribution by Hour

This graph illustrates the distribution of crimes across different hours of the day. Such information helps identify peak crime hours, which may indicate periods when criminal activities are more likely to occur.

3.4 Arrest Distribution

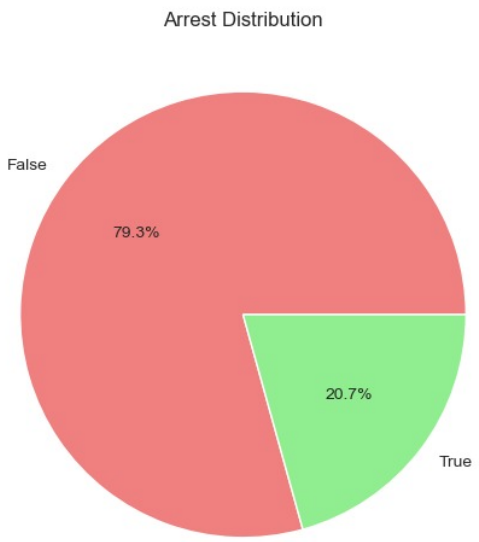

Figure 4: Arrest Distribution

The arrest distribution chart shows the proportion of crimes that resulted in an arrest compared to those that did not. This information helps understand the effectiveness of law enforcement actions.

3.5 Crime Density Heatmap

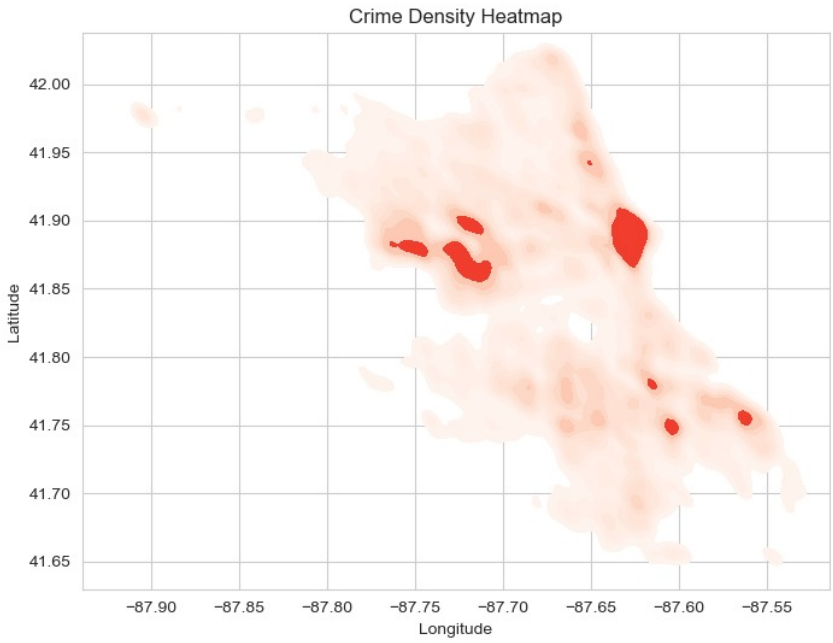

Figure 5: Crime Density Heatmap

The heatmap visualizes the spatial concentration of crimes across geographic locations. Areas with higher densities indicate potential crime hotspots.

4 Methodology

The methodology followed in this project involves several stages, including data preprocessing, feature engineering, hotspot labeling, feature selection, and model training.

4.1 Data Preprocessing

Data preprocessing is a crucial step in preparing the dataset for machine learning. This stage involves handling missing values, converting date columns into a datetime format, and removing records with incomplete location information. The dataset is also cleaned to ensure that all features are in a suitable format for analysis.

4.2 Feature Engineering

Feature engineering involves creating meaningful input variables that improve the performance of machine learning models. In this project, several temporal features are extracted from the date column, including hour of occurrence, month, and day of the week. These features help capture temporal patterns in crime occurrences.

Spatial features such as latitude and longitude are also used to represent the geographical location of crimes. Additional features such as arrest status and community area provide further contextual information.

4.3 Hotspot Label Creation

To identify crime hotspots, the geographical region is divided into grid cells based on latitude and longitude coordinates. The number of crimes occurring in each grid cell is calculated. Areas with crime counts above a certain threshold are labeled as hotspots. This threshold is determined using statistical analysis of the crime count distribution.

4.4 Feature Selection

Feature selection is performed using the ANOVA statistical test. This method evaluates the relationship between each feature and the target variable, allowing the model to focus on the most informative features. Selecting the most relevant features improves model performance and reduces computational complexity.

5 Machine Learning Models

Several machine learning models are implemented and evaluated in this study.

5.1 Random Forest

Random Forest is an ensemble learning method that constructs multiple decision trees during training. Each tree produces a prediction, and the final output is obtained by aggregating the predictions of all trees. Random Forest is known for its robustness and ability to reduce overfitting.

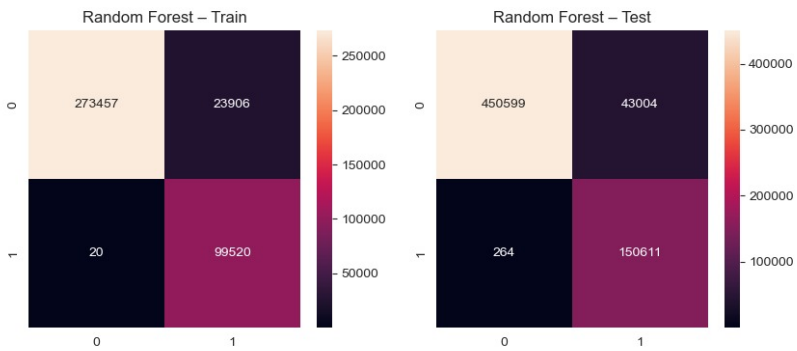

Figure 6: Random Forest Confusion Matrix

5.2 XGBoost

XGBoost is a gradient boosting algorithm that builds models sequentially by correcting errors made by previous models. It is widely used for structured data problems due to its high efficiency and predictive accuracy.

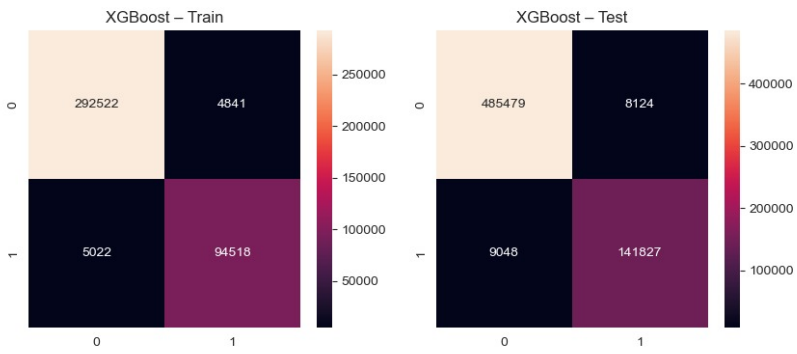

Figure 7: XGBoost Confusion Matrix

5.3 LightGBM

LightGBM is a gradient boosting framework designed for speed and efficiency. It uses a leaf-wise tree growth strategy, which allows it to achieve high accuracy while maintaining fast training times.

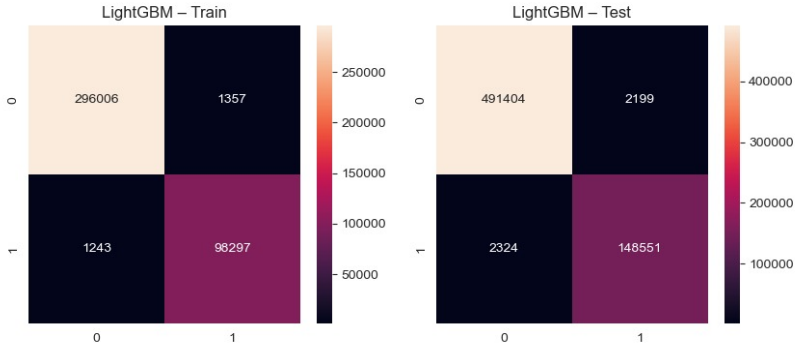

Figure 8: LightGBM Confusion Matrix

5.4 CatBoost

CatBoost is a machine learning algorithm specifically designed to handle categorical variables effectively. It applies ordered boosting techniques to reduce prediction bias and improve model accuracy.

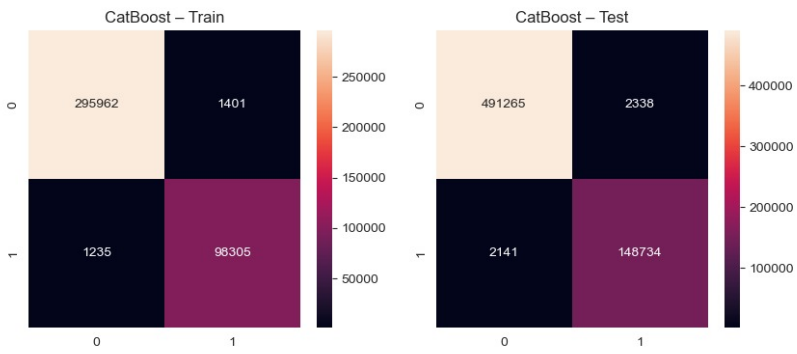

Figure 9: CatBoost Confusion Matrix

6 Model Evaluation

To evaluate model performance, several metrics are used, including accuracy, precision, recall, and F1-score. These metrics provide a comprehensive understanding of how well each model performs in predicting crime hotspots.

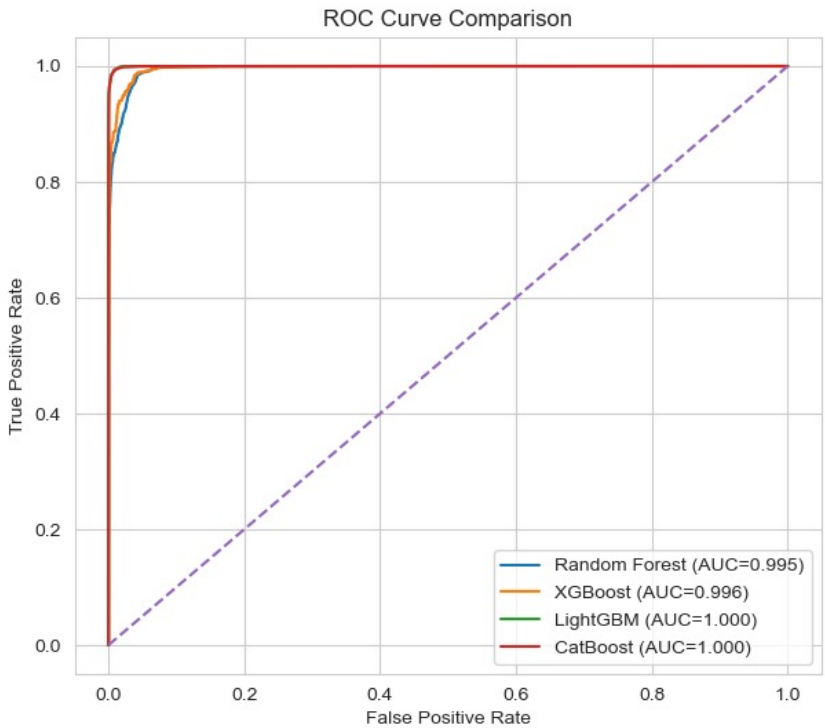

Figure 10: ROC Curve Comparison

The ROC curve compares the performance of different models by analyzing their true positive and false positive rates.

7 Model Performance Comparison

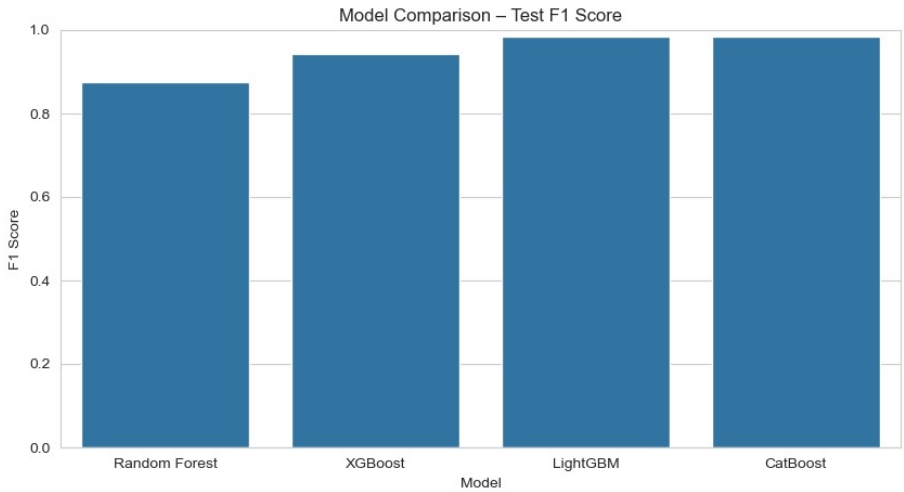

Figure 11: Model Comparison

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========== FINAL MODEL COMPARISON ==========
           Model  Train Accuracy  Test Accuracy  Precision    Recall  F1 Score
0  Random Forest        0.939718       0.932863   0.777889  0.998250  0.874400
1        XGBoost        0.975150       0.973355   0.945822  0.940030  0.942917
2       LightGBM        0.993449       0.992982   0.985413  0.984597  0.985005
3       CatBoost        0.993359       0.993050   0.984524  0.985809  0.985166
```

Figure 12: Final Model Performance Table

The comparison results show the relative performance of all models based on the selected evaluation metrics.

8 Conclusion

This project demonstrates how machine learning models can be used to analyze crime datasets and predict potential crime hotspots. By leveraging spatial and temporal features, the models can identify patterns that indicate regions with a higher likelihood of criminal activity.

Among the evaluated models, gradient boosting algorithms such as XGBoost and LightGBM demonstrate strong predictive performance. These models are capable of capturing complex relationships within the dataset and producing accurate predictions.

The proposed approach can assist law enforcement agencies in implementing predictive policing strategies. By identifying high-risk areas in advance, authorities can allocate resources more effectively and take preventive actions.

Future work may involve integrating additional datasets such as demographic information, weather conditions, and socio-economic indicators to further enhance the accuracy of crime prediction models.